

LAN8710A MII PHY Customer Evaluation Board

Assy 6583
PCB Revision C
Schematic Revision 1.2

Design Details

Board:
Assy 6583

Chip:
SMSC LAN8710A

Board Form Factor:
MII

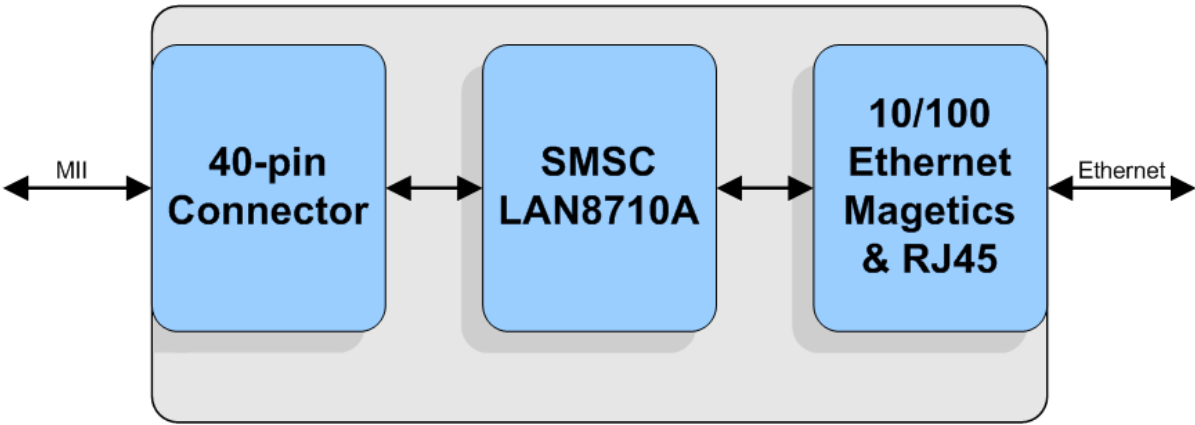
Assembly:
32 Lead QFN w/ Exposed GND Pad

Circuit Diagrams utilizing SMSC Products Are Included As A Means Of Illustrating Typical Semiconductor Applications: Consequently Complete Information Sufficient For Construction Purposes Is Not Necessarily Given. The Information Has Been Carefully Checked And Is Believed To Be Entirely Reliable. However, No Responsibility Is Assumed For Inaccuracies. Furthermore, Such Information Does Not Convey To The Purchaser Of The Semiconductor Devices Described Any License Under The Patent Rights Of SMSC Or Others. SMSC Reserves The Right To Make Changes At Any Time In Order To Improve Design And Supply The Best Product Possible.

Revision History

- Rev 1.0:**
Initial release, Rev C
- Rev 1.1:**
All Pages - Changed from LAN8710 to LAN8710A
- Rev 1.2:**
C19 changed from 0.1uF to 470pF.

BLOCK DIAGRAM



ITEM	Page(s)
Title Page	1
LAN8710A / MII / Magnetics	2

[illegible]

REGOFF

LED1/REGOFF 1 R40 2 LED1 ANODE

3 0

R38 4.75K 1/10W

LED1/REGOFF 1 R41 2 LED1 CATHODE

3 330

3-Pad Resistor Default Population Settings: 1~2

nINTSEL

LED2/nINTSEL 1 R42 2 LED2 ANODE

3 0

R39 4.75K 1/10W

LED2/nINTSEL 1 R43 2 LED2 CATHODE

3 330

SIGNAL	POPULATE	DEPOPULATE
PHYAD[0] = 0	R37	R24
PHYAD[1] = 0	R36	R25
PHYAD[2] = 0	R35	R26
PHYAD[0] = 1	R24	R37
PHYAD[1] = 1	R25	R36
PHYAD[2] = 1	R26	R35
MODE[0] = 0	R31	R30
MODE[1] = 0	R32	R29
MODE[2] = 0	R33	R28
MODE[0] = 1	R30	R31
MODE[1] = 1	R29	R32
MODE[2] = 1	R28	R33
(R)MI Select	R34	Not Supported
Internal 1.2V Reg. Enabled	R40, R41: 1-2	R40, R41: 2-3
Internal 1.2V Reg. Disabled	R40, R41: 2-3	R40, R41: 1-2
Interrupt Function Enabled on nINT/TXER/TXD4 Signal	R42, R43: 1-2	R42, R43: 2-3
Interrupt Function Disabled on nINT/TXER/TXD4 Signal	R42, R43: 2-3	R42, R43: 1-2

VDDCR

TP3

Test Point - Green
VDDCR
DNP

C19
470pF
50V
5%

C20
1.0uF
16V
10%

Note:
C19 & C20 decoupling capacitors for VDDCR should be placed as close to LAN8710A as possible.